

The Feynman Lectures on Physics, Volume II

Problem Set 4

Due 2022.5.08

1 The Laws of Induction (5pt)

Consider two coaxial loops of radius a which are separated by a distance d ; $d \gg a$. A current $I = K_0 t^2$ is sent through one loop (1), as shown in Fig. 1; the resistance of the other loop (2) is R .

- (a) Neglecting self-inductance, what is the torque τ on loop (2)?
- (b) Show that, if the self-inductance is neglected, the force on loop (2) is

$$F = \frac{24\pi^4 a^8 K_0^2 t^3}{(4\pi\epsilon_0 c^2)^2 d^7 R}$$

- (c) In what direction is the force?
- (d) In what way (qualitatively) is the true force and torque different from your estimate; i.e., how does the self-inductance affect the torque and force?
- (e) Explain what would happen in parts (a) and (b) above if loop (2) were rotated 90° about an axis normal to the common axis of the loops.

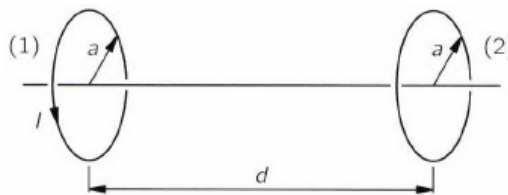


Figure 1:

2 The Maxwell Equations (6pt)

Prove **Alfven's theorem**: In a perfectly conducting fluid (say, a gas of free electrons), the magnetic flux through any closed loop moving with the fluid is constant in time. (The magnetic field lines are, as it were, “frozen” into the fluid.)

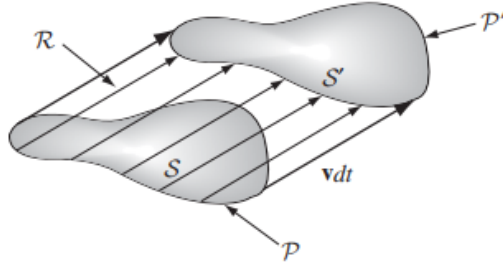


Figure 2:

- (a) Use Ohm's law together with Faraday's law, to prove that if $\sigma = \infty$ and $\mathbf{J}$ is finite, then

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B})$$

- (b) Let $\mathcal{S}$ be the surface bounded by the loop ($\mathcal{P}$) at time t , and $\mathcal{S}'$ a surface bounded by the loop in its new position ($\mathcal{P}'$) at time $t + dt$ (see Fig. 2). The change in flux is

$$d\Phi = \int_{\mathcal{S}'} \mathbf{B}(t + dt) \cdot d\mathbf{a} - \int_{\mathcal{S}} \mathbf{B}(t) \cdot d\mathbf{a}$$

Use $\nabla \cdot \mathbf{B} = 0$ to show that

$$\int_{\mathcal{S}'} \mathbf{B}(t + dt) \cdot d\mathbf{a} + \int_{\mathcal{R}} \mathbf{B}(t + dt) \cdot d\mathbf{a} = \int_{\mathcal{S}} \mathbf{B}(t + dt) \cdot d\mathbf{a}$$

(where $\mathcal{R}$ is the "ribbon" joining $\mathcal{P}$ and $\mathcal{P}'$), and hence that

$$d\Phi = dt \int_{\mathcal{S}} \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} - \int_{\mathcal{R}} \mathbf{B}(t + dt) \cdot d\mathbf{a}$$

(for infinitesimal dt). Use the method of Sec. 7.1.3 in Griffith's book to rewrite the second integral as

$$dt \int_{\mathcal{P}} (\mathbf{B} \times \mathbf{v}) \cdot d\mathbf{l}$$

and invoke Stokes' theorem to conclude that

$$\frac{d\Phi}{dt} = \int_{\mathcal{S}} \left(\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) \right) \cdot d\mathbf{a}$$

Together with the result in (a), this proves the theorem.

3 The Principle of Least Action (4pt)

The action for a charged particle in electromagnetic field can be written as

$$S = - \int_{t_1}^{t_2} mc^2 \sqrt{1 - v^2/c^2} + q\phi - q\mathbf{v} \cdot \mathbf{A}$$

Please use the principle of least action to derive the equation of motion for a charged particle in electromagnetic field.